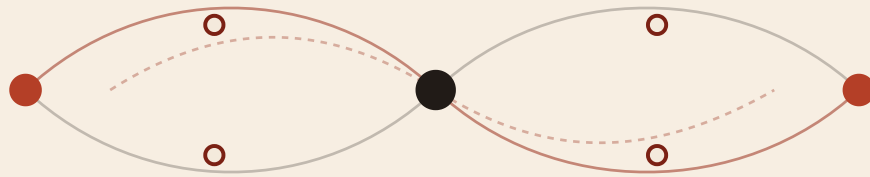


A High-Level View of Work Context Infrastructure



1. From perception to intention

All work begins, at least in some form, with an idea about something that could be different. Someone observes something in their life or in the reality around them and perceives a possibility, a need, a problem, or a condition worth preserving. From that perception, an objective begins to take shape.

The objective does not appear in isolation. Perception is situated. People encounter reality through lenses influenced by their experience, knowledge, biological constitution, social position, and cultural context. These factors do not mechanically determine one conclusion, but they influence what becomes visible, which distinctions seem important, and which possibilities can be imagined.

The system does not need to model human perception itself. It needs to preserve the attributable transition from a situated perception to a declared and revisable objective.

At first, the objective often appears as a single monolithic intention: one large, still formless whole with little internal detail or clarity. A person may sense a direction before they can describe the destination precisely. They may know that something should change without yet knowing the work, constraints, or sequence required to change it. The intention carries a purpose, but its parts, relationships, and much of its meaning remain implicit.

3. The problem the system must solve

The central purpose of the proposed system is to preserve the relationship between local work and the larger contexts that explain it.

A local piece of work should remain connected to:

- the objective from which it ultimately emerged;
- the interpretations, assumptions, and evidence that shaped that objective;
- the Plans, discoveries, specifications, and decisions that refined it;
- the authority that permitted a person, agent, or tool to act;
- the attempts, effects, and artifacts through which the work became concrete; and
- the evidence used to judge whether the result advanced anything larger.

This is not only a task-management problem. A task manager can record assignments while losing their justification. It is not only a document-management problem. Documents can preserve text while leaving their semantic and operational relationships implicit. It is not only a workflow or multi-agent problem. A workflow can execute the wrong route correctly.

The deeper problem is the loss of context between levels of work.

Every piece of work should remain connected to the larger contexts that give it purpose, meaning, and legitimate authority.

The system must preserve five independently inspectable paths: **purpose**, **authority**, **assignment**, **causation**, and **realization**. They may converge on the same outcome, but none can be inferred from another. An execution can, for example, be authorized without having been assigned to the purpose-bearing task. The remainder of this essay explains the properties, responsibilities, and validation strategy required to preserve these paths.

When inspected, a path may be witnessed, missing, conflicted, or superseded. These are explanatory distinctions rather than a fixed response vocabulary.

Reading map

- [Sections 1–16](#) establish the problem and the properties the system may need.
- [Sections 17–30](#) describe the candidate shape and its responsibilities.
- [Sections 31–34](#) name preservation and formalization questions.
- [Sections 35–36](#) define the first discriminating test.
- [Sections 37–38](#) preserve open questions, ownership gaps, and limits.

Proposal: why familiar framings may be partial

Proposal status. The following table presents a scope hypothesis introduced by this essay, not a finding inherited from an authoritative source. It proposes that work-context infrastructure should not be identified exclusively with any one of these familiar system categories. Each category may still contribute part of the implementation; the proposed limitation is that, on its own, it does not guarantee preservation of purpose, authority, assignment, causation, and realization across the path from intention to effect. Accepting this table would accept that scope hypothesis, not select an architecture or reject these systems as components.

Framing	What it contributes	Proposed limitation if used alone
Task-management system	Assignment, scheduling, status, and coordination.	Does not by itself guarantee preservation of justification, authority, or outcome evidence.
Document repository	Durable artifacts, versions, and retrieval.	Does not by itself make semantic relationships or operational consequences explicit.
Workflow engine	Repeatable transitions and execution coordination.	Usually operates after meaning, authority, and acceptance boundaries have been defined.
Multi-agent orchestrator	Dispatch, coordination, and agent execution.	Covers an execution mechanism, not necessarily the full lineage from intention to accepted effect.

FIVE INDEPENDENTLY INSPECTABLE PATHS

01

Purpose

why the work exists

02

Authority

what permits action

03

Assignment

who carries
responsibility

04

Causation

what produced
change

05

Realization

what supports the
outcome

CANDIDATE OUTCOME

4. Moving from the part to the whole

From any task, attempt, artifact, or effect, a person should be able to move upward through its context:

```
Why does this exist?  
Which work asked for it?  
Which Plan or route produced that work?  
Which objective does the route serve?  
Which decision authorized this action?  
Which assumptions and evidence shaped those choices?
```

This upward path checks purpose and authority. It makes orphan work visible. A task with no defensible upward path may still be technically clear, but the system cannot explain why it should be performed.

The path need not be one fixed chain. A task may acquire its purpose through one set of relations and its authority through another. What matters is that the relevant paths remain inspectable and that the system does not invent missing links.

5. Moving from the whole to its parts

From an objective, a person should also be able to move downward:

```
Which Plans attempt to realize this objective?  
Which work is active, blocked, abandoned, or complete?  
Which decisions changed the route?  
Which tasks and attempts are operating now?  
Which effects and artifacts were produced?  
What evidence indicates that the objective is being advanced?
```

17. A recursive grammar of work

A familiar progression offers a useful first reading:

```
Intent
-> Plan
-> Research
-> Discovery / Design
-> Specification
-> Implementation
-> Verification
```

It is not one irreversible pipeline. Research may challenge a Plan. Verification may reopen a Specification. A local implementation question may require another Plan. Work kinds can recur at different scopes and participate in several contexts.

The property to preserve is not one universal order. It is that the applicable route, dependencies, bypasses, evidence, authority, and reopenings remain explicit and historically reconstructable.

18. Decomposition without loss of lineage

Complex work must be decomposed into bounded assignments. Each assignment should receive an immediate objective, selected context, constraints, an expected result, applicable gates, and an authority boundary.

The assignment should not need to carry the whole project in its prompt or working memory. It should retain a resolvable path to the larger purpose while receiving only the context relevant to its responsibility.

```
context available through lineage
!= context materialized for one execution
```

This distinction allows the broader context to remain preserved without making every local worker interpret everything.

31. Candidate invariants

Before choosing a kernel, the system can name properties it may need to preserve:

- names, paths, classifications, and projections do not silently redefine identity;
- accepted history remains reconstructable;
- relationships preserve type, direction, provenance, scope, and applicable version;
- derived paths retain their direct witnesses and composition rules;
- projections cannot manufacture facts or authority;
- lineage does not automatically delegate tools, evidence, budget, or effect permission;
- cheap descriptions do not silently become types, truths, or permissions;
- revocation and supersession remain visible;
- checker acceptance does not imply physical enforcement; and
- local validity does not automatically imply global compatibility.

These are candidate invariants. Naming them does not prove that they are necessary, sufficient, or enforced.

35. The smallest useful vertical slice

The first system should test one end-to-end question:

Given one candidate outcome, observed artifact, or observed effect, can the system first determine whether accepted terminal evidence exists and then reconstruct defensible purpose, authority, assignment, and causal execution paths without treating adjacency as proof?

The purpose and authority paths converge but do not collapse:

```
purpose:      objective -> chosen route -> local work
authority:    accepted permission -> authorized execution -> attempt
assignment:   local work -> bounded responsibility -> attempt
causation:    attempt -> observed contribution -> candidate outcome
realization:  local work -> supported outcome claim -> candidate outcome
```

The first slice should test whether an explicit assignment bridge is required. Convergence on the same outcome supplies a counterexample candidate when an authorized attempt was carrying a different responsibility: the test should determine whether the execution basis must connect that attempt to the purpose-bearing local work rather than relying on a shared outcome alone. The necessity and sufficiency of that bridge remain open under `OD-06`.

Terminal evidence separates two obligations. A **terminal-state witness** shows that an attempt or effect route reached an applicable terminal condition. An **outcome-binding witness** connects that terminal activity to the candidate outcome. The first without the second is activity evidence, not causal or realization evidence. Likewise, finalized content without an accepted producer binding is durable content rather than realization evidence, and an intended effect without an accepted outcome observation remains a request.

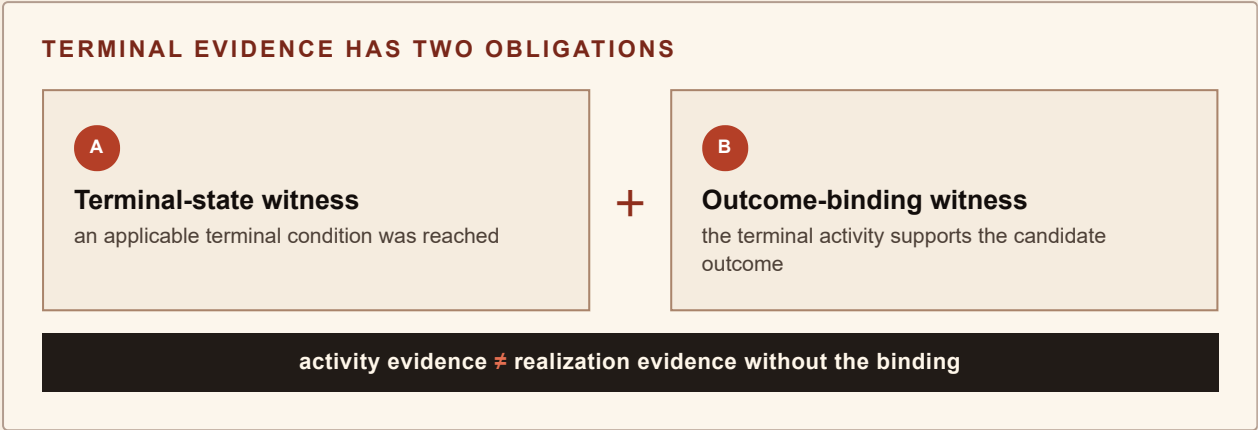
For any accepted causal binding, the slice should expose its attributable source, the acceptance occurrence or process, the scope and version in which it applies, and resolvable supporting evidence. Which process may accept such a binding and what evidence is sufficient remain open.

The slice should support traversal in both directions, expose missing links instead of filling them, and preserve historical changes to each path. It should not require a universal ontology, a final trusted-boundary design, or formalization of every relation before testing the core value.

Current repository correspondence for the first slice

The present repository specifies two non-canonical correspondence variants. For runtime-managed flows, ACI specifies ownership of confirmed Dispatch authority, and APT specifies the corresponding `aci_managed` authority-snapshot reference. For legacy-managed flows, ACI specifies that authority

remains with the legacy ledger, while APT specifies a `legacy_ledger` external-owner reference rather than duplicated authority. These names describe current repository contracts; they do not select the general architecture, establish a future contract, or prove that a runtime exists.



37. Open questions

- Which things deserve stable identities, and at what point?
- Which relationships are direct facts, and which may be derived?
- What makes an upward path sufficient to justify local work?
- What evidence shows that micro work advanced a macro objective?
- When does changed context invalidate earlier authority?
- How much context should a leaf worker receive?
- Can an object participate in several Plans or projects without ambiguous authority?
- Which candidate invariants are global, domain-local, or configurable?
- What must be checked by validators, formal methods, empirical tests, or human judgment?
- How should independently governed contexts compose?
- How can maintenance cost and recovered value be measured?

System-view result

- **Status:** flag — complete as a companion essay, but its open decisions and ontology terms do not yet have instantiated downstream owners.
- **Target boundary:** macro-to-micro work context from human objective through local effect, including purpose, authority, evidence, change, candidate architecture, and a first vertical slice.
- **Stakeholder altitude:** begins with a general reader and increases toward product and architecture stakeholders; schema and runtime implementation are deferred.
- **Shape:** perception -> intention -> decomposition -> context preservation -> authority and evidence -> architecture responsibilities -> invariants and formalization -> first slice.
- **Layering:** human situation and current work are given; context selection, decomposition, governance, and projections are optimized; accepted history accumulates.
- **Open decisions named:** six, all explicitly unassigned.
- **Decided-nothing check:** pass — the named stances remain open.
- **Term-deferral check:** pass with dependency — provisional navigation terms are explicitly deferred to unassigned future ontology work.
- **Evidence boundary:** repository-backed problem and current mechanisms; proposed synthesis for the full architecture and validation route.